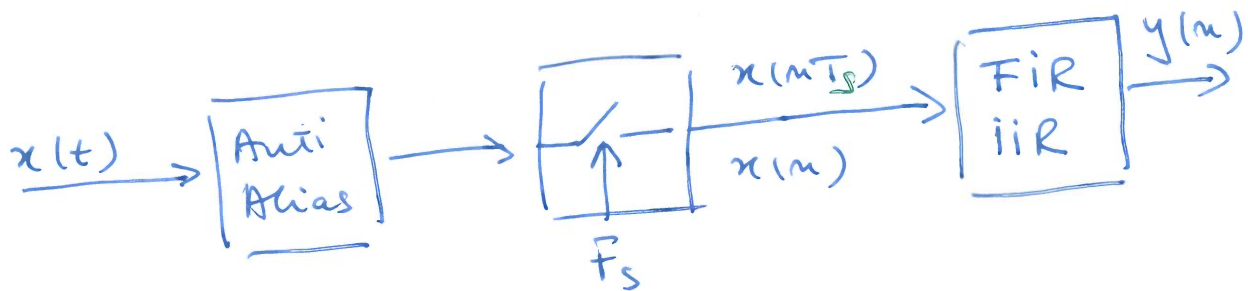


Subjects - DIGITAL FILTERS - z-transformFIR - Finite Impulse ResponseIIR - Infinite Impulse ResponseFIR EXAMPLE

$$y(n) = \frac{1}{2}x(n) + \frac{1}{2}x(n-1)$$

difference equation

The z transform

$$Y(z) = 0.5X(z) + 0.5X(z)z^{-1}$$

X(z)

$$\frac{Y(z)}{X(z)} = 0.5 + 0.5z^{-1} = H(z)$$

Transfer Function

$$= 0.5(1 + z^{-1})$$

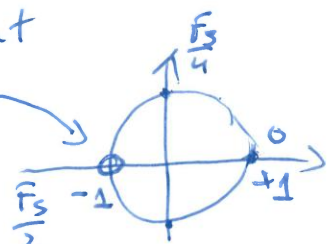
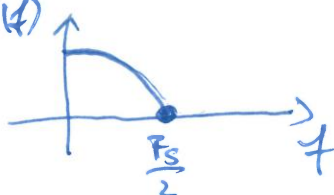
→ z transform of FIR

$$= 0.5 \frac{z+1}{z}$$

pole at 0 does not count

Has a zero at $z = -1$ ⇒ at $\frac{F_s}{2}$

⇒



11

iiR

(2)

$$x(n) \quad x(n-1) \quad \dots \quad x(0)$$

$$n \rightarrow \infty$$

$$\underline{y(n)} = x(n) + x(n-1) + x(n-2) + \dots + x(0) \quad \Bigg| \quad \frac{1}{n}$$

$$y(n-1) = x(n-1) + x(n-2) + \dots + x(0)$$

$$y(n) = \underline{y(n-1)} + x(n)$$

$\downarrow z$

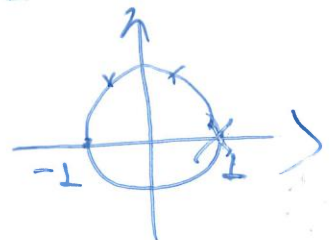
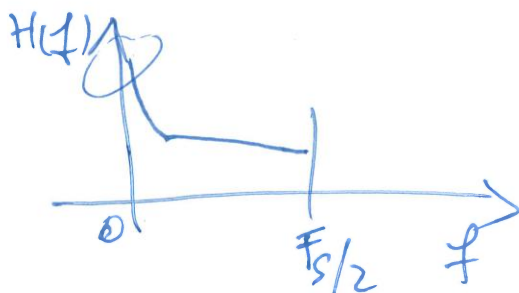
$$Y(z) = Y(z) \cdot z^{-1} + X(z) \quad \Bigg| \quad X(z)$$

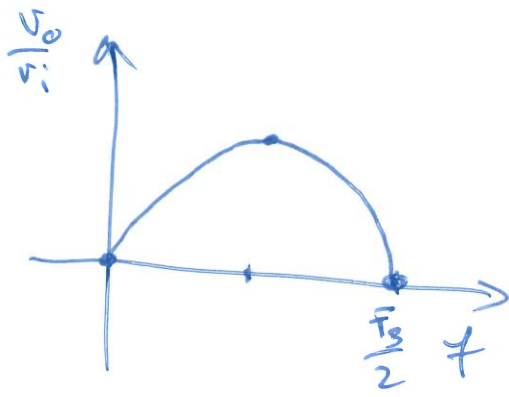
$$\cancel{H(z)} = \frac{Y(z)}{X(z)} = \frac{Y(z)}{X(z)} z^{-1} + 1 \quad \Bigg| \quad -\frac{Y(z)}{X(z)} z^{-1}$$

$$\frac{Y(z)}{X(z)} (1 - z^{-1}) = 1 \quad \Bigg| \quad 1 - z^{-1}$$

$$\frac{Y(z)}{X(z)} = \frac{1}{1 - z^{-1}} \quad \Bigg| \quad \frac{z}{z} = \frac{z}{z - 1}$$

$$z \cdot z^{-1} = 1$$





FIR

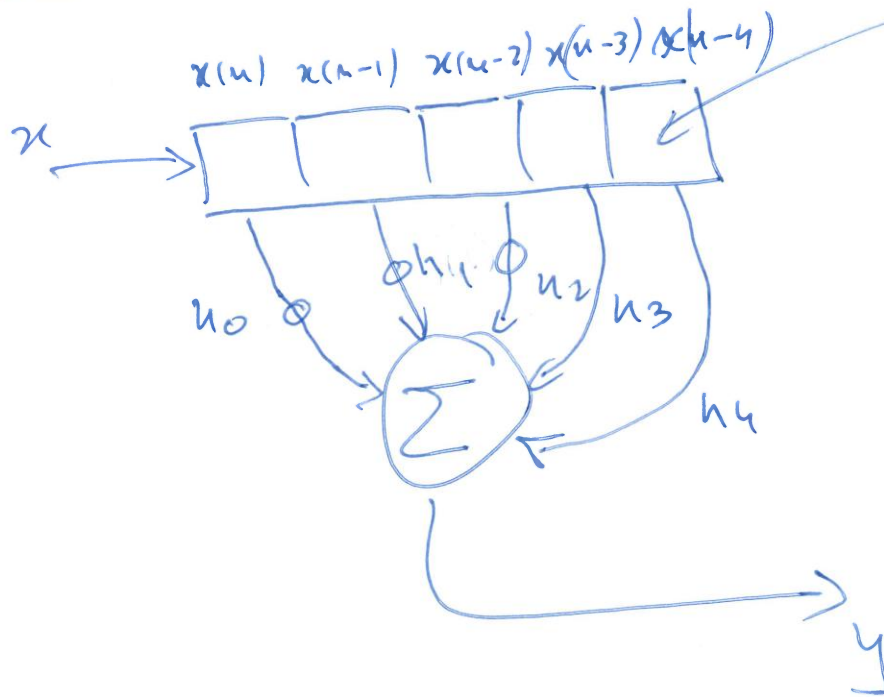
$$y(n) = b_0 x(n) + b_1 x(n-1) + b_2 x(n-2) + \dots + b_{M-1} x(n-M+1)$$

IIR

$$y(n) = a_1 y(n-1) + a_2 y(n-2) + \dots + a_M y(n-M) + b_0 x(n) + b_1 x(n-1) + \dots + b_{M-1} x(n-M+1)$$

FIR

$M=5$



Direct
IMPLEMENTATION

```
int x[5];
```

```
int a;
```

```
a = next_sample;
```

```
for (i=0; i<5; i++) {  
    x(i+1) = x(i);  
}
```

```
x(0) = a;
```

```
int ym = 0;
```

```
for (i=0; i<5; i++) {  
    ym = ym + x(i);  
}
```